



FEYMAN

AHRS User Guide

V1.0



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1. Introduction

FEYMAN (FMI) IMU modules (IM23TCB-AHRS, INS240, MCS10 series etc) support multiple operating modes. This document describes the software configuration steps required to operate the IMU in AHRS (Attitude and Heading Reference System) mode.

In AHRS mode the IMU computes Roll, Pitch, and Yaw (heading) using its internal MEMS accelerometers, gyroscopes, and magnetometer, with no GNSS signal required. This makes AHRS mode suitable for indoor environments, and any application where GNSS is unavailable.

After completing the steps in this guide the IMU will output Roll, Pitch, and Yaw data continuously.

2. AHRS Setup Instruction

In order for the IMU to operate in AHRS mode, a different AHRS firmware version is needed, starting with MCS10.

Example: **Version:MCS10xxx**

2.1 Updating the Bootloader

The bootloader is the low-level program that manages firmware upgrades. It must be updated first to ensure compatibility with the AHRS firmware.

Step 1, Send the bootloader update command

In the GNSS&MEMS Tool Debug terminal, send:

AT+UPDATE_BOOT

After sending **AT+UPDATE_BOOT** the IMU enters bootloader update mode and waits for the file upload.

Step 2, Upload the bootloader file

- In the GNSS&MEMS Tool, navigate to the firmware/bootloader upload section.
- Click Open and select the bootloader file provided by FEYMAN.
- Click Upload and wait for the transfer to complete.

Demo video: https://youtu.be/bB1tS1u-Y_c

2.2 Updating the Firmware

After the bootloader is updated, the main application firmware must be flashed using the following two-command sequence.

Step 1, Reset the system

Send the following command to reset the IMU:

AT+SYSTEM_RESET

Step 2, Enter firmware update mode

Within **3 seconds** of the system reset, send:

AT+UPDATE_APP



NOTICE: AT+UPDATE_APP must be sent within 3 seconds of AT+SYSTEM_RESET. If the window is missed, repeat both commands from Step 1.

Step 3, Upload the AHRS firmware file

- In the GNSS&MEMS Tool, navigate to the firmware upload section.
- Click Open and select the AHRS firmware file provided by FEYMAN.
- Click Upload and wait for the process to complete. Do not disconnect the IMU during the upload.

Demo video: https://youtu.be/bB1tS1u-Y_c

2.3 Enabling AHRS Output

With the firmware updated, configure the IMU to output AHRS data (message definition in APPENDIX A) by sending:

AT+AHRS_OUTPUT=UART1,ON

This enables the AHRS message stream on UART1. The IMU will begin outputting Roll, Pitch, and Yaw data continuously.

Command parameters

Parameter	Description
UART1	Specifies the output serial port.
ON	Enables AHRS output. Replace with OFF to disable.

2.4 Calibrating the IMU

Magnetometer calibration corrects installation related errors such as distortions caused by the mounting position, nearby metallic components, or electrical interference. This step is critical for accurate Yaw (heading) output.

NOTICE: Perform calibration after the IMU is installed in its final mounting position. Moving or reorienting the device after calibration may require re-calibration.

Step 1, Start calibration

Send the following command to begin the magnetometer calibration (APPENDIX B) process:

AT+MAG_CALI_MANU=START

Once calibration starts, slowly rotate the IMU through all orientations in 3D space (roll, pitch, and yaw) to allow the magnetometer to sample the full magnetic field.

Recommended rotation motion

- Rotate slowly around the X-axis (roll, tilt left and right)
- Rotate slowly around the Y-axis (pitch, tilt forward and backward)



- Rotate slowly around the Z-axis (yaw, spin horizontally through 360°)

Step 2, Save calibration

Once the rotation sequence is complete, save the calibration data:

AT+MAG_CALI_MANU=SAVE

The calibration data is stored in the IMU and will persist across power cycles. The IMU applies the calibration automatically on startup.

After completing the above steps the IMU will output Roll, Pitch, and Yaw data in AHRS mode without requiring GNSS.

Demo video: <https://youtu.be/f2aN9qJ5t00>



3. AHRS Commands

AT+UPDATE_BOOT	Enter bootloader update mode.
AT+SYSTEM_RESET	Perform a full system reset.
AT+UPDATE_APP	Enter firmware update mode.
AT+AHRS_OUTPUT=UARTX,ON/OFF	AHRS (Roll/Pitch/Yaw) output.
AT+MAG_CALI_MANU=START	Begin manual magnetometer calibration.
AT+MAG_CALI_MANU=SAVE	Save the magnetometer calibration parameters.
AT+WORK_MODE=2	Set work mode to AHRS mode.
AT+SAVE_ALL	Save all current configuration parameters. Persists across power cycles.
AT+VERSION	Query the firmware version installed on the IMU.

4. Troubleshooting

Symptom	Possible Cause / Resolution
Cannot connect to IMU	Verify the correct COM port is selected. Ensure no other application is using the port. Try a different cable.
AT+UPDATE_APP fails or times out	The command must be sent within 3 seconds of AT+SYSTEM_RESET. Retry both commands quickly in sequence.
Firmware upload fails	Verify the firmware file is the correct version for your hardware model. Check the USB connection and retry.



APPENDIX A

Communication Protocol

1. AHRS message

Index	Message content	
1	Message header	0X66
2		0X6D
3		0X69
4		0X41
5	Message length	0X0020
7	Millisecond	uint32_t
11	Acceleration X	uint16_t
13	Acceleration Y	int16_t
15	Acceleration Z	int16_t
17	Gyroscope X	int16_t
19	Gyroscope Y	int16_t
21	Gyroscope Z	int16_t
23	Magnetometer X	int16_t
25	Magnetometer Y	int16_t
27	Magnetometer Z	int16_t
29	Roll	int16_t
31	Pitch	int16_t
33	Yaw	int16_t
35	Status	uint32_t
39	CRC16	int16_t

Annotation:

The conversion factor of the acceleration output: 8g / 32000.0

Conversion factor for angular velocity output: 500°/s / 32000.0

The conversion factor of the Angle output: 180° / 32000.0



2. Magnetometer Calibration Message

Index	Message content	
1	Message header	0X66
2		0X6D
3		0X69
4		0X62
5	Message length	length
7	Millisecond	uint32_t
11	Calibration indicator 1	float
15	Calibration indicator 2	float
19	Calibration indicator 3	float
23	Calibration indicator 4	float
27	Calibration indicator 5	float
31	Calibration indicator 6	float
35	Calibration indicator 7	float
39	Calibration indicator 8	float
43	Calibration residual	float
47	CRC16	uint16_t

Annotation:

Calibration indicator 1,it needs to be less than 5 at the end of the calibration.
Calibration indicator 2,it needs to be less than 5 at the end of the calibration.
Calibration indicator 3,it needs to be less than 5 at the end of the calibration.
Calibration indicator 4,it needs to be less than 0.05 at the end of the calibration.
Calibration indicator 5,it needs to be less than 0.05 at the end of the calibration.
Calibration indicator 6,it needs to be less than 0.05 at the end of the calibration.
Calibration indicator 7,it needs to be less than 0.02 at the end of the calibration.
Calibration indicator 8,it needs to be less than 0.05 at the end of the calibration.

Calibration is successful if and only the following conditions are met:

```
Calibration_success = indicator 1 < 5  
                    && indicator 2 < 5  
                    && indicator 3 < 5  
                    && indicator 4 < 0.05  
                    && indicator 5 < 0.05  
                    && indicator 6 < 0.05  
                    && indicator 7 < 0.02  
                    && indicator 8 < 0.05;
```



3. CRC16 Rules

Calculate CRC16 from 7th byte to the last byte as follow:

```
unsigned short poly = 0x1021;

uint8_t* data = buffer + 6;
uint16_t* crc = (uint16_t*)(buffer + offset);
*crc = 0;

while (length --)
{
    *crc ^= (*(data++) << 8);
    for(uint8_t i = 0; i < 8; i++)
    {
        if(*crc & 0x8000) *crc = (*crc << 1) ^ poly;
        else *crc = *crc << 1;
    }
}
```